Nika Ilieva

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PROFESSIONAL SUMMARY

Device designer and engineer developing solutions for Duke Health systems. Experienced in diagnosing client needs and translating into technical implementation and regulatory-aware product development.
Portfolio: *https://nikailieva335-sudo.github.io*

EDUCATION

Duke University Durham, NC
BSE, Biomedical Engineering. GPA: 3.47/4.0. *Expected December 2026*
Fellowships: BME Design Fellows (program pairing students with Duke Health clinicians to design medical devices)
Pratt Research Fellowship ($6,000 award for independent research)

EXPERIENCE

Design Fellow Intern, Durham, NC
BME Design Fellows Internship *May 2026 – Present*

- Conducted structured discovery meetings with Duke Children's Hospital, Duke Eye Center, and the Duke Center for Global Women's Technologies to identify clinical needs, define user requirements, produced user needs and specifications documentation.
- Designing a wearable CO2 monitoring device for neuromuscular dystrophy patients targeting under $300, reducing cost burden by over 70%. Designed the device schematic and PCB, developed firmware and built testable MVP prototypes.
- Trained and optimized ML classification models using Neuton.AI across 5 iterative approaches to improve diagnostic accuracy. Resolved failure modes including class imbalance and systematic misclassification. Achieved 89% Type B recall (+22.3pts vs. baseline), 98% specificity in the final model, and a 53% reduction in false positives between model 4 and 5. Built an embedded inference validation pipeline.
- Developed a real-time insertion-angle guidance system for the mHealth Tympanometer: implemented embedded firmware (Zephyr RTOS, nRF52833) using IMU accelerometer and gyroscope data fusion to detect improper probe angles in real time and issue a warning.

Engineer Consultant, Durham, NC
DesignHub at Duke Innovation Co-Lab *February 2026 – Present*

- Provided on-demand design consultations for clinical and research partners across Duke Pediatrics, Duke Children's Hospital, Duke Medical Physics, Duke Surgery, and Duke Lemur Center; diagnosed the underlying need behind each client request before recommending solutions.
- Designed two EMG comfort toy prototypes for Duke Pediatric Clinic: a faithful device reproduction and a frog-character redesign; client adopted the frog design upon evaluation.
- Identified that a client's proposed injection-molding approach for a pelvic anatomy phantom was technically incompatible with their mesh scan data and less accurate than alternatives; recommended switching to the Stratasys J750 Digital Anatomy printer, improving fidelity of the model.

3D Printing Consultant and Operator, Durham, NC
Bluesmith at Duke Innovation Co-Lab *February 2025 – Present*

- Consulted on 30+ projects per semester, advising clients on printer selection, materials, and manufacturing strategy for biomedical and clinical research; translated vague problem statements into actionable plans accounting for budget, FDA and regulatory, and research constraints.
- Operate and maintain MJP2500, Formlabs 3/4B, Formlabs Fuse, Stratasys F370, and Stratasys J750 Digital Anatomy printers across SLA, SLS, FDM, and PolyJet technologies.

Researcher, Author, and Speaker, Durham, NC
BEETL (Biomedical Engineering Education Teaching Lab), Duke University *January 2025 – Present*

- Identified a systemic gap in engineering leadership education; conducted a scoping literature review (150+ articles screened and analyzed) quantified the problem finding 37% of leadership curricula were optional and 40% of literature lacked a clear definition of engineering leadership. Presented findings as LEAD speaker at the 2026 ASEE Annual Conference and as a speaker at The Ohio State University's Conference on Civic Thought, Education and Leadership.
- Co-led redesign of BME260 (Modeling Cellular and Molecular Systems to integrate leadership/followership curriculum

SKILLS

Medical Device and Prototyping: **Zephyr RTOS, C/C++, KiCad, CAD (Onshape), 3D printing (SLA, SLS, FDM, PolyJet), 3DSlicer**
AI and Data Analysis: **Python (NumPy, pandas, Matplotlib, SciPy, scikit-learn, Torch, Transformers), ML, Edge AI**
Software Development: **Git/GitHub, CI/CD, pytest, Flask, RESTful APIs, MongoDB, Tkinter, pycryptodome, PEP-8**
Client Communication: **Background research, Discovery Interviews, Microsoft Powerpoint, Canva, Technical Communication**
Research and Automation: **PyHamilton robotics, automation workflows, Benchling, protein engineering, molecular biology**